

Screening

1. Introduction

The objective of this project is to build and evaluate machine learning models that predict

whether a chemical compound is biologically active or inactive against a target. The task is

formulated as a binary classification problem, where 1 represents an active compound and 0

represents an inactive compound.

This type of prediction is relevant for virtual screening in drug discovery. In a real workflow,

computational models can help prioritize promising compounds before laboratory testing. The goal

of this project is not to claim that a new drug candidate has been discovered, but to compare

different machine learning models for activity prediction and analyze their behavior.

2. Dataset

The dataset used in this project is the Kaggle Drug Discovery Virtual Screening Dataset:

<https://www.kaggle.com/datasets/shahriarkabir/drug-discovery-virtual-screening-dataset>

The project code repository is available at:

https://github.com/milicajeftic/drug_discovery_ml

The raw dataset contains 2,000 compounds and 17 columns. These include:

14 numeric modeling features

2 identifier columns: 'compound_id' and 'protein_id'

1 target column: 'active'

The target variable is moderately imbalanced:

Inactive compounds: approximately 70%

Active compounds: approximately 30%

Three columns contain missing values:

'logp'

'polar_surface_area'

'hydrophobicity'

Each of these columns has 60 missing values. Because the missing values do not occur in exactly

the same rows, removing rows with any missing value leaves 1,826 complete samples.

3. Preprocessing

The preprocessing workflow is implemented in 01_edata_preprocessing.ipynb.

The main preprocessing steps are:

1. Remove rows with missing values.

2. Remove identifier columns from the feature matrix.

3. Separate features from the active target variable.

4. Create stratified train/validation/test splits using a 70/15/15 ratio.

5. Fit StandardScaler only on the training set.

6. Transform validation and test sets using the training-fitted scaler.

4. Models

Logistic Regression

Logistic Regression was used as the baseline model. It is simple, interpretable, and provides a

useful reference point for more complex models. Balanced class weights were used to account for

the moderate class imbalance.

Random Forest

Random Forest was used as a non-linear ensemble model. It can capture feature interactions and

provides feature importance scores. Balanced class weights were used during training.

Gradient Boosting

Gradient Boosting was used as another tree-based ensemble method. Balanced sample weights were

used because the model does not directly support `class_weight`.

Neural Network

A feed-forward neural network was implemented using `sklearn.neural_network.MLPClassifier`. This

model uses two hidden layers, ReLU activation, L2 regularization, early stopping, and balanced

sample weights. This satisfies the neural network component proposed in the project while

remaining compatible with the project .conda environment.

5. Evaluation Metrics

All models were evaluated using the same metrics:

? Accuracy

? Precision

? Recall

? F1-score

? ROC-AUC

Recall is especially important in virtual screening because false negatives represent active

compounds that the model fails to identify.

Learning curves were also analyzed using F1-score to compare training and cross-validation performance as the training set size increases. They show that Random Forest and Gradient Boosting

remain consistently strong. Logistic Regression improves gradually with more data, and the Neural

Network is more variable on smaller training subsets.

6. Results

Final test-set performance:

Model	Accuracy	Precision	Recall	F1-score	ROC-AUC
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Logistic Regression	0.9891	0.9651	1.0000	0.9822	0.9998
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Random Forest	1.0000	1.0000	1.0000	1.0000	1.0000
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Gradient Boosting	1.0000	1.0000	1.0000	1.0000	1.0000
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Neural Network	0.9854	0.9647	0.9880	0.9762	0.9993
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The best-performing models are Random Forest and Gradient Boosting. Both achieved perfect

validation and test performance on the current processed dataset.

7. Error Analysis

The test-set error analysis shows:

Model	False Positives	False Negatives	Total Errors
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Logistic Regression	3	0	3
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Random Forest	0	0	0
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Gradient Boosting	0	0	0
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Neural Network	3	1	4
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Random Forest and Gradient Boosting made no test-set errors. Logistic Regression made three false

positive errors but no false negatives, which is favorable in a virtual screening context because

it did not miss active compounds. The Neural Network made four errors, including one false

negative.

8. Interpretability

Feature importance analysis was performed for the tree-based models.

The most important Random Forest features were:

1. binding_affinity

2. logp_pi_interaction

3. logp

For Gradient Boosting, binding_affinity dominated the feature importance almost completely.

This suggests that the activity_label is strongly related to binding affinity and closely related

descriptors. This also explains why the tree-based models achieved perfect performance.

9. Limitations

The main limitation of this project is related to the dataset. The dataset appears to be synthetic

and highly separable. Since binding_affinity is included as an input feature and is strongly

associated with the target label, the classification task may be easier than a real-world virtual

screening problem.

In real drug discovery, binding affinity may not always be known before prediction. If the

activity label was derived from binding affinity, using binding affinity as an input feature can

overestimate model performance.

Therefore, the results should be interpreted as a comparison of model behavior on this dataset,

not as proof that the same performance would be achieved on real experimental drug discovery data.

Another limitation is that only a limited hyperparameter search was performed. The model

parameters were selected manually and kept fixed during the final comparison. Since the models

already achieved very high validation and test performance, an extensive grid search was not

necessary for the main goal of comparing model families, but it could be added in future work.

10. Conclusion

This project successfully developed and evaluated a machine learning pipeline for predicting

compound activity in a virtual screening setting. The workflow includes data checking,

preprocessing, baseline modeling, tree-based modeling, neural network modeling, model comparison,

and error analysis.

The tree-based models performed best, with Random Forest and Gradient Boosting achieving perfect

test-set results. Logistic Regression also performed very strongly and served as a reliable

baseline. The Neural Network achieved high performance but did not outperform the tree-based

models.

The final conclusion is that tree-based ensemble models are the best choice for this dataset.

However, the dominant role of binding_affinity should be discussed as an important limitation when

interpreting the results.